

CS471 – Block 1 Master Solutions

Lessons 4-8: Search, Heuristics, and Adversarial Decisions

Block 1 Equations & Concepts Reference

- **Evaluation Function ($f(n)$):** *The total estimated cost of the cheapest path through node n . $g(n)$ is the exact cost to reach n from the start, and $h(n)$ is the heuristic estimated cost from n to the goal.*

$$f(n) = g(n) + h(n)$$

- **Admissibility Condition:** *A heuristic is admissible if it never overestimates the true cost (h^*) to reach the goal. Guarantees optimality in A^* Tree Search.*

$$0 \leq h(n) \leq h^*(n)$$

- **Consistency Condition:** *The estimated cost from n to the goal is no greater than the step cost to a successor n' plus the estimated cost from n' to the goal. Guarantees optimality in A^* Graph Search.*

$$h(n) \leq c(n, a, n') + h(n')$$

- **Dominance:** *A dominant heuristic is more accurate and guarantees A^* will expand fewer or equal nodes. If $h_2(n) \geq h_1(n)$ for all n (and both are admissible), then h_2 strictly dominates h_1 .*
- **Minimax Algorithm:** *Models a zero-sum adversarial game where the MAX agent tries to maximize the utility score while the MIN opponent tries to minimize it.*

$$\max_a \min_{a'} \dots$$

- **Expectimax Algorithm:** *Models stochastic environments where chance nodes dictate outcomes. Instead of assuming worst-case scenarios, the agent maximizes the expected (average weighted) utility.*

$$\max_a \sum_{s'} P(s') V(s')$$

Part I: Guided Practice

Lesson 4. Problem Formulation

An autonomous Combat Search and Rescue (CSAR) helicopter must navigate a 10x10 coordinate grid to reach a downed pilot. Some grid squares contain hostile anti-air threats and cannot be traversed.

- (a) Formulate this search problem by explicitly defining the **State Space**, **Initial State**, **Actions**, **Transition Model**, **Goal Test**, and **Path Cost**.

Solution

- **State Space:** All valid (x, y) coordinates on the 10x10 grid that do not contain threats.
- **Initial State:** The starting (x, y) coordinate of the CSAR helicopter.
- **Actions:** Move North, South, East, West (assuming no diagonal capability).
- **Transition Model:** Taking action a in state (x, y) returns the resulting adjacent (x', y') coordinate.
- **Goal Test:** Does the current (x, y) equal the downed pilot's (x, y) ?
- **Path Cost:** 1 unit of fuel/time per grid square traversed.

- (b) What is the difference between a "World State" and a "Search State" in this scenario?

Solution

The **World State** includes every physical detail of the environment (e.g., wind speed, rotor temperature, precise enemy radar frequencies). The **Search State** is an abstracted model that retains *only* the variables mathematically required to plan a path to the goal (the x, y coordinates and threat boundaries).

Lesson 5. Uninformed Search

A cyber-agent maps a compromised network. From the entry router (S), it discovers nodes A and B. A connects to C and D. B connects to E. The goal is to locate the adversary's backdoor at node E.

- (a) If the agent uses **Breadth-First Search (BFS)**, in what exact order will the nodes be expanded? What data structure manages the BFS frontier?

Solution

BFS explores layer by layer.

Order: S, A, B, C, D, E.

BFS manages its frontier using a **FIFO Queue** (First-In, First-Out).

- (b) If the agent uses **Depth-First Search (DFS)**, in what exact order will the nodes be expanded? What data structure manages the DFS frontier?

Solution

DFS dives down the left-most path first.

Order: S, A, C, D, B, E.

DFS manages its frontier using a **LIFO Stack** (Last-In, First-Out).

Lesson 6 & 7: Informed Search and Heuristics

- (a) Execute **Greedy Best-First Search**. List the sequence of nodes expanded. What is the total cost of the resulting path?

Solution

Greedy search expands the node with the lowest $h(n)$.

- Start at S ($h = 6$). Successors: A ($h = 4$), B ($h = 1$).
- Expand B (lowest h). Successor: G ($h = 0$).
- Expand G (Goal found).

Nodes expanded: S $\rightarrow$ B $\rightarrow$ G.

Path Cost: $g(S \rightarrow B) + g(B \rightarrow G) = 5 + 2 = 7$.

- (b) Execute **A* Search**. Calculate $f(n) = g(n) + h(n)$ for the frontier at each step. List the final path and its total cost.

Solution

A* expands the lowest $f(n) = g(n) + h(n)$.

- **Step 1:** Expand S. Frontier: A ($f = 2 + 4 = 6$), B ($f = 5 + 1 = 6$). Tie, expand A.
- **Step 2:** Expand A. Frontier: B (from S, $f = 6$), C ($f = (2 + 3) + 3 = 8$), B (from A, $f = (2 + 2) + 1 = 5$).
- **Step 3:** Expand B (via A, $f = 5$). Frontier: G ($f = (4 + 2) + 0 = 6$), C ($f = 8$).
- **Step 4:** Expand G. (Goal found).

Nodes expanded: S $\rightarrow$ A $\rightarrow$ B $\rightarrow$ G.

Path Cost: $2 + 2 + 2 = 6$. (A* found a cheaper path than Greedy).

- (c) Is the provided heuristic $h(n)$ admissible? Prove your answer by comparing $h(n)$ to the true lowest cost $h^*(n)$ for nodes A and B.

Solution

Yes, it is admissible. It never overestimates the true cost to the goal.

For A: $h(A) = 4$. True cost $h^*(A)$ via B is $2 + 2 = 4$. ($4 \leq 4$)

For B: $h(B) = 1$. True cost $h^*(B)$ is 2. ($1 \leq 2$)

Because $h(n) \leq h^*(n)$ for all nodes, the heuristic is admissible.

Lesson 8: Adversarial Decisions

- (a) Perform the Minimax algorithm. Write the backed-up values inside the MIN nodes and the root MAX node. Which action (Left or Right) should MAX choose?

Solution

- Left MIN node chooses the minimum of its children: $\min(4, 8) = 4$.
- Right MIN node chooses the minimum of its children: $\min(2, 9) = 2$.
- Root MAX node chooses the maximum of the MIN nodes: $\max(4, 2) = 4$.

MAX should choose the **Left** action.

- (b) If the algorithm evaluates nodes from left to right, using **Alpha-Beta Pruning**, which leaf node(s) will be pruned (not evaluated)? Explain the mathematical logic that triggers the prune.

Solution

The leaf node with value 9 will be pruned.

Logic: After evaluating the Left MIN node, MAX knows it can guarantee a score of at least $\alpha = 4$. When evaluating the Right MIN node, the first leaf reveals a value of 2. The MIN node will therefore return a value *at most* 2. Since $2 < 4$, MAX will never choose the Right path. The algorithm prunes the remaining children of the Right node because they cannot possibly affect the root decision.

Part II: Independent Student Practice

Practice 1: Problem Formulation

An AI agent manages power routing for an orbital satellite. The satellite has 3 solar panels that can independently be ON or OFF. The agent's goal is to turn all 3 panels ON. The agent can only toggle one panel per action.

- (a) Define the size of the state space.

Solution

There are 3 panels, each with 2 possible states (ON/OFF).

$$\text{State Space Size} = 2^3 = \mathbf{8} \text{ total states.}$$

- (b) Define the exact Goal Test.

Solution

Goal Test: Do panels $(P_1 == \text{ON}) \wedge (P_2 == \text{ON}) \wedge (P_3 == \text{ON})$?

Practice 2: Search Algorithms

An intelligence routing algorithm navigates a priority tree. The starting node is 1. Its children are 2 and 3. Node 2's children are 4 and 5. Node 3's child is 6.

- (a) If the algorithm manages its frontier using a **LIFO Stack** (Depth-First Search), what is the exact expansion order to reach node 5?

Solution

DFS explores the deepest left-most nodes first.

Expansion Order: $1 \rightarrow 2 \rightarrow 4 \rightarrow \mathbf{5}$.

Practice 3 & 4: A* Search and Heuristics

A loyal wingman drone must navigate a grid. Moving vertically or horizontally costs 1 unit. Moving diagonally costs 1.4 units. The drone is at coordinate (0,0) and the target is at (2,2).

- (a) The agent uses the **Manhattan Distance** heuristic. Calculate $h(0,0)$.

Solution

Manhattan distance is $|x_2 - x_1| + |y_2 - y_1|$.

$$h(0,0) = |2 - 0| + |2 - 0| = 2 + 2 = \mathbf{4}$$

- (b) Is the Manhattan Distance heuristic admissible in this specific environment? Mathematically prove your answer.

Solution

No, it is not admissible.

Because diagonal movement is allowed at a cost of 1.4, the true lowest cost $h^*(0, 0)$ to reach $(2, 2)$ is taking two diagonal steps: $1.4 + 1.4 = 2.8$.

The heuristic estimates $h(0, 0) = 4$. Because $4 \not\leq 2.8$, the heuristic *overestimates* the true cost ($h(n) > h^*(n)$). Therefore, it is inadmissible and will break A*'s optimality guarantee.

Practice 5: Stochastic Decisions (Expectimax)

An Electronic Warfare (EW) AI is deciding whether to Jam a radar or Stay Silent. If it Jams, there is a 60% chance the adversary radar reboots (Utility = 10) and a 40% chance the adversary escalates power (Utility = -5). If it Stays Silent, the utility is definitively 2.

- (a) Calculate the expected value (EV) of the "Jam" action.

Solution

$$\begin{aligned} EV(\text{Jam}) &= P(\text{Reboot}) \times U(\text{Reboot}) + P(\text{Escalate}) \times U(\text{Escalate}) \\ &= (0.60)(10) + (0.40)(-5) \\ &= 6 - 2 \\ &= 4 \end{aligned}$$

- (b) According to the Expectimax algorithm, which action should the EW AI take?

Solution

The expected value of Jamming is 4. The known value of Staying Silent is 2. Since $4 > 2$, the agent will choose the **Jam** action to maximize its expected utility.